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Exploring AI/ML in Power Electronics - Component Selection and Converter Topology Optimizations

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APEC 2026

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PREES Laboratory
NC STATE UNIVERSITY
PACKAGING RESEARCH IN ELECTRONIC ENERGY SYSTEMS

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1 of 22



The Power Electronics Problem

- As each area of ... the electronics industry evolves, it often is accompanied by the development of a succession of new circuits
- Each new circuit indeed appears different, employing different components in differing configurations, and claims an assortment of distinct features...
- Without a considerable investment of laboratory time to construct, evaluate, and compare each candidate circuit, it usually is difficult to realistically appraise the relative merits of one approach over another.

FRED C. LEE, 1985*

*Now we are entering the Software era of
AI/ML Design and the Digital Twin*

*Preface to Modern Dc-to-Dc Switchmode Power Converter Circuits, Severns & Bloom

Micro vs Power Electronics - A Fundamental Difference

- Microelectronics, i.e. Data/Telecom, compete in “information processing performance” with high throughput as higher performance.
 - Microelectronics can agree on the physical systems, at least supporting common materials and process advancements. Their competition is in the *ETHEREAL* world.
- Power-Electronics compete in “energy performance” AND at zero size, zero weight, zero loss and zero cost.
 - Power-electronics **cannot agree** on the physical systems, since each application requires a different mix of Electrical/Mechanical/Thermal performance with a plethora of electro-physical topologies, materials and processes. Competition is in the *PHYSICAL* world.

The complexity for power electronics DESIGN creates an immense data set that requires an AI/ML approach

*Preface to Modern Dc-to-Dc Switchmode Power Converter Circuits, Severns & Bloom

Outline

*Presented is an AI/ML review of foundational works.
Many newer refinements have been developed since this
presentation was first proposed with many more evolving daily.*

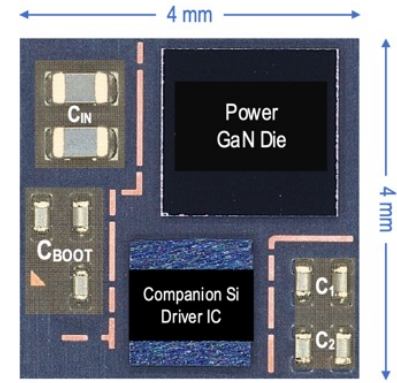
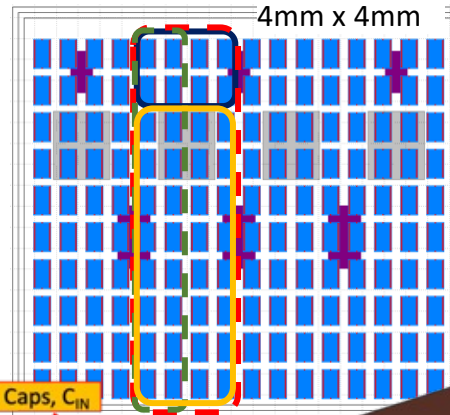
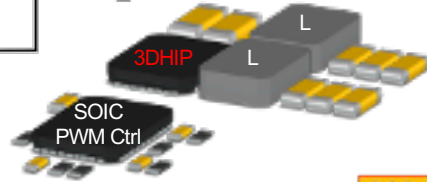
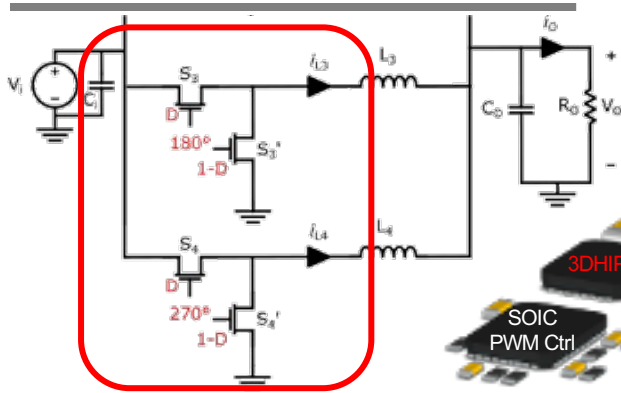
Seven works were analyzed as foundational works exploring use of AI in PE circuit design thru automating and optimizing complex tasks.

AI tools for component selection and optimization analyze vast datasets of

- specifications, pricing, and component availability, and
- evaluate and balance multiple circuit parameters to suggest configurations that achieve the best possible performance.

A simple AI/ML Optimization is given as a short tutorial, and crosses into the microelectronics world for power delivery

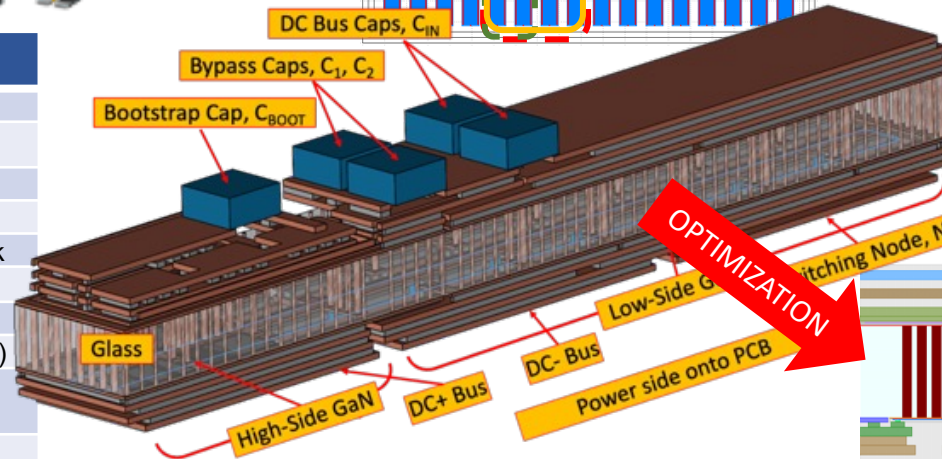
(Short tutorial) Simple ML Optimization



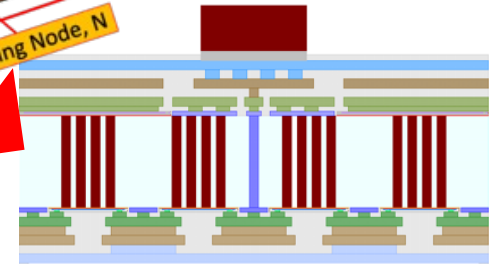
"A 32A 5V-Input, 94.2% Peak Efficiency High-Frequency Power Converter Module Featuring Package-Integrated Low-Voltage GaN NMOS Power Transistors," 2021 Symp on VLSI Circuits, 978-4-86348-780-2

Exemplar Design Requirements (Intel)

Parameters	Values
V_i (Input DC Bus)	5V
V_o (Output)	1V
Efficiency	> 90%
Type	Interl'd Sync Buck
No. of Phases	4
Duty	0.21
I_o (average)	32A (8A per phase)
Peak L Ripple Current (per ϕ)	6.4A (20% of I_o)
V_o ripple	0.1V (10% of V_o)



Half symmetry: 1-ph of a 4-ph interleaved Buck showing passives and power terminals

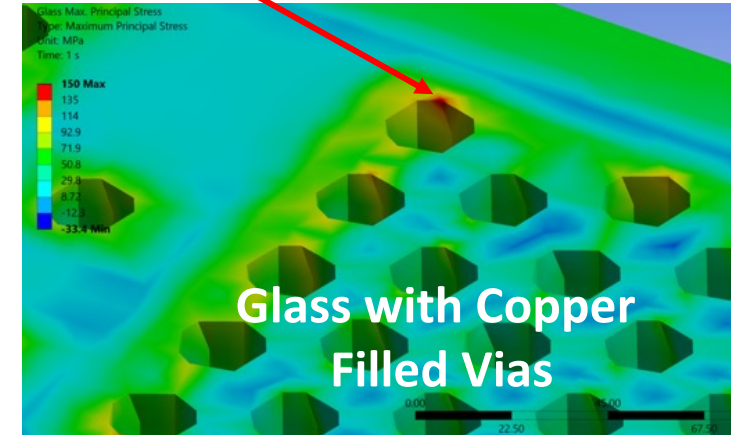
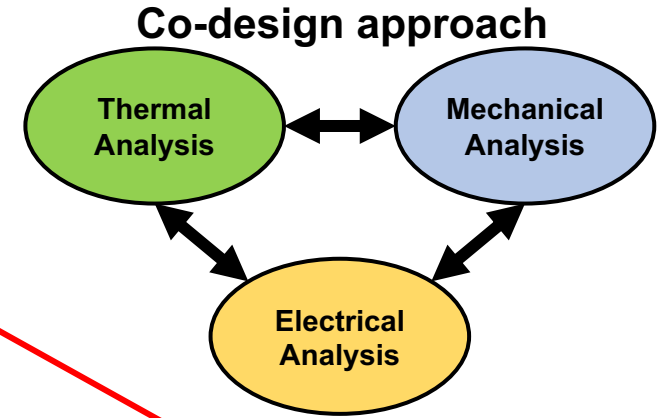


Problem Statement (Cu/TGV)

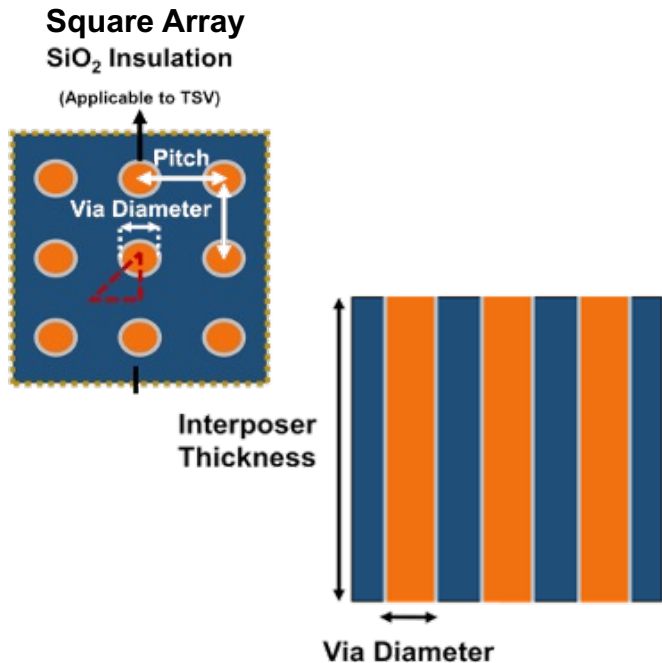
- The Cu in the Through Glass Via expands at a higher rate. The max prin. stress in glass was 150MPa vs. rupture modulus of 135MPa.

Objective

- Developing a multi-objective **optimization** framework incorporating **Online Artificial Neural Network (O-ANN)** to create a layout for optimum performance.
- AI/ML replaces the calculation loop, freeing the engineer to focus on architecture and innovation.



Design Space



$$AR = \frac{\text{Interposer Thickness}}{\text{Via Diameter}}$$

Design parameter	Description
Aspect ratio	2, 3, ..., 9, 10
Pitch (μm)	$2d, 2.2d, \dots, 3.8d, 4d$
Interposer thickness (μm)	100
TSV SiO ₂ thickness (μm)	1
Interposer material	Glass and Silicon
Via array topology	Square and hexagonal

Parameters: Aspect Ratio,
Pitch, Material, Array Topology

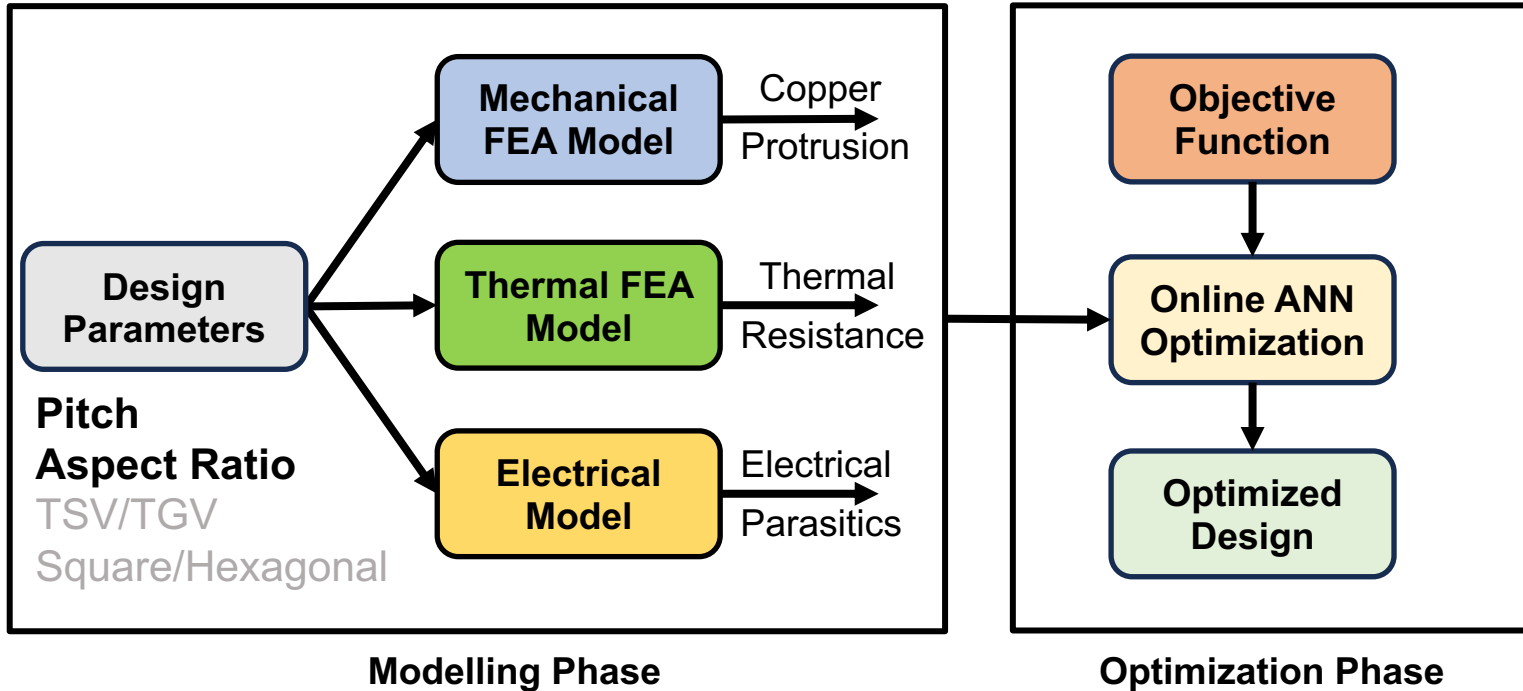
2, 3, ..., 9, 10

Data Set: $2d, 2.2d, \dots, 3.8d, 4d$

Silicon and glass

Square and hexagonal

Modeling & Optimization Phases Using ML



Objective Function

Mechanical

Thermal

Electrical

$$F(x) = w_1 \left(\frac{u_z}{u_z^\circ} \right) + w_2 \left(\frac{R_{th}}{R_{th}^\circ} \right) + w_3 \left[\frac{1}{4} \left(\frac{R_{el}}{R_{el}^\circ} + \frac{L_s}{L_s^\circ} + \frac{L_m}{L_m^\circ} + \frac{C_m}{C_m^\circ} \right) \right]$$

Mechanical Constraints: $\begin{cases} g_1(y): \sigma_{P,sub} - \sigma_{u,sub} < 0 \\ g_2(y): \sigma_{P,SiO_2} - \sigma_{u,SiO_2} < 0 \\ g_3(y): \sigma_{Cu} - \sigma_{y,Cu} < 0 \end{cases}$

Penalty Function

$$+ r \sum_{i=1}^3 \langle g_i(y) \rangle^2$$

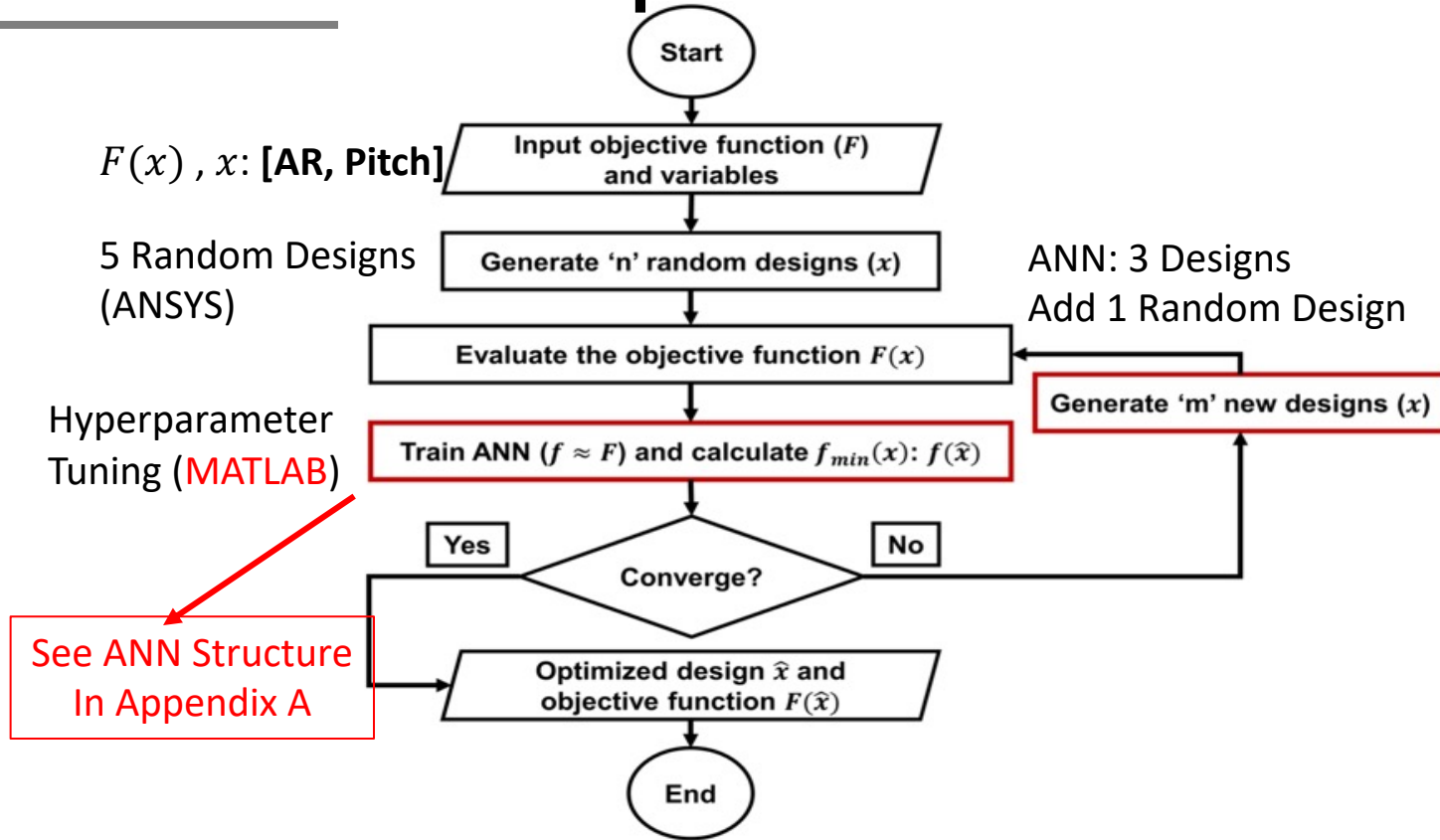
$$\sum_{i=1}^3 w_i = 1, \quad \langle g_i(y) \rangle = \max(0, g_i(y)), \quad r = \frac{16F(x_1)}{n}$$

Optimization Weighting Coefficients

Case	w_1	w_2	w_3	Description
I	1/3	1/3	1/3	Equal preference
II	5/7	1/7	1/7	Mechanical preference
III	1/7	5/7	1/7	Thermal preference
IV	1/7	1/7	5/7	Electrical preference

$F(x)$	Objective function
x	Design variable vector
w_i	Weight coefficients
R_{th}	Thermal resistance
R_{el}	Electrical resistance
L_s, L_m	Self & mutual inductance
C_m	Mutual capacitance
u_z	Copper protrusion
$\circ$	Maximum value of parameter
r	Penalty function
$g_i(y)$	Mechanical constraints
σ_P, σ_u	Principal & ultimate stress
σ_y	Yield stress
x_1	Initial random design
n	Number of design variables

Online ANN-Based Optimization

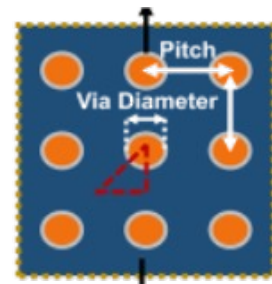
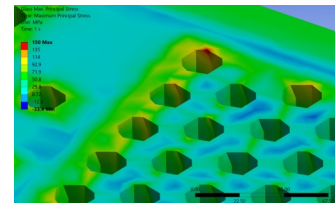


Results

TGV Optimization Results

* I: Equal preference, II: Mechanical preference, III: Thermal preference, and IV: Electrical preference

Structure	Case*	Pitch (xd)	AR	u_z (nm)	R_{th} (K/W)	R_{el} (m Ω)	L_s (pH)	L_m (pH)	C_m (fF)
Square TGV	I	2	6	13.0	0.0130	34.5	42.2	150	29.7
	II	2	10	7.8↓	0.0130	65.4	50.6	206	29.4
	III	2	6	13.0	0.0130↓	34.5	42.2	150	29.7
	IV	2.8	2	39.8	0.0253	6.05↓	27.1↓	47.2↓	23.3↓



- Mechanically optimized** → Max. 50% protrusion reduction compared to equal preference.
- Electrically optimized** → Up to 23% capacitance, 82% electrical resistance reduction, 36% self, and 69% mutual inductance reduction compared to equal preference.
- Thermally optimized** → Up to 49% thermal resistance reduction versus electrical preference.

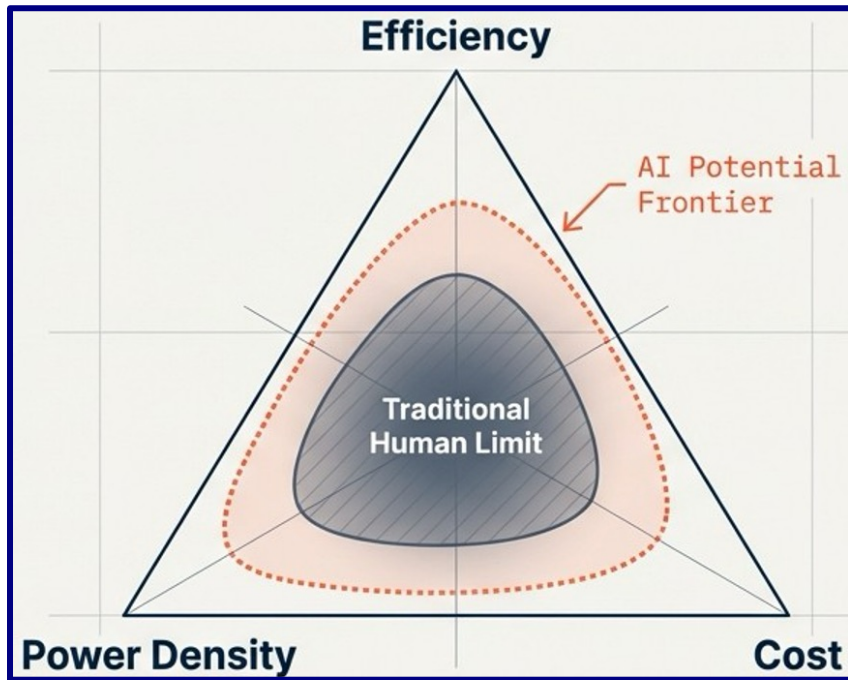
Maximum Z-direction displacement (Optimized)

AI/ML For Power Electronics

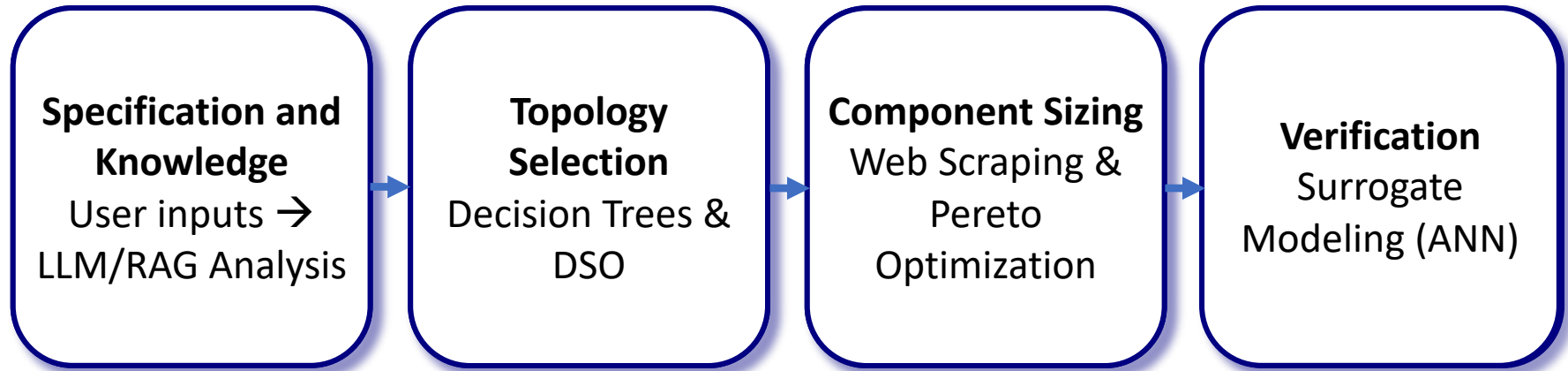
Engineering Bottleneck – Why Intuition Scales Poorly

- **Human Expertise:** Relies on rules of thumb and misses non-intuitive optima.
- **Simulation Costs:** Physics-based sims (Spice/Ansys) take minutes-hours per iteration
- **Data Overload:** Databases can contain >100,000 parts (e.g., 110k capacitors), making manual search impossible.

THE POWER CONVERTER TRADE-OFF TRIAD



The Integrated AI/ML Design Workflow



Manual Process : Days/Weeks

AI Process: Sub-second Inference

Large Language Model (LLM), Retrieval-Augmented Generation (RAG) , Design Space Optimization (DSO)

The “AI Consultant” (Start of your agent)



User Query

e.g. “High step-down ratio required”

LLM + RAG Framework



Retrieval
Augmented
Generation (RAG)

Academic
Literature/IEEE
Database

Output

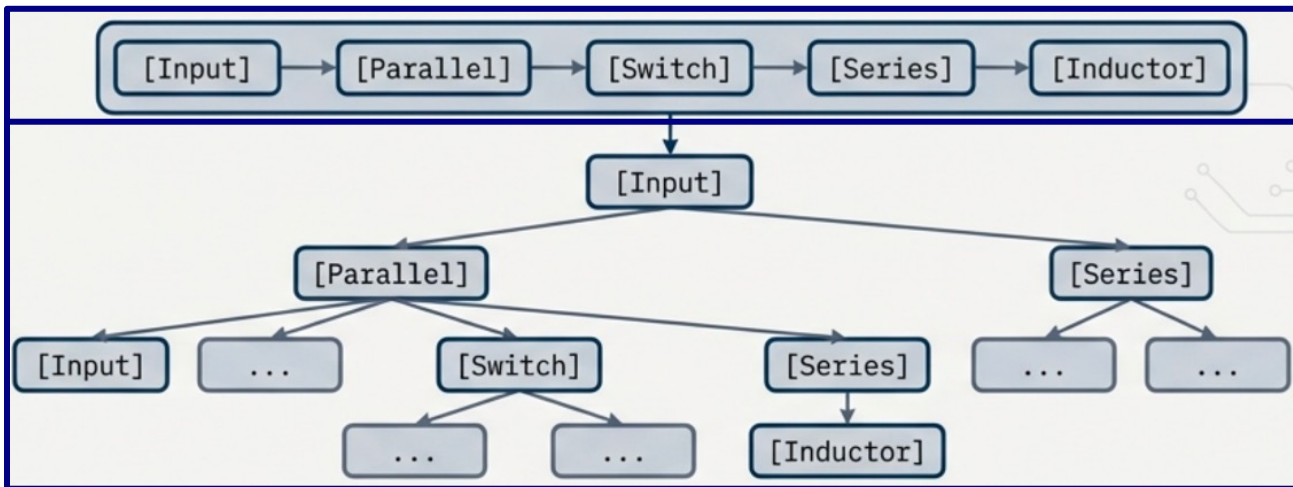
Recommendation

Recommendation: LLC
Converter

Justification: Higher
efficiency compared to
DAB

Benefit: Limiting AI to verified engineering literature prevents hallucination.

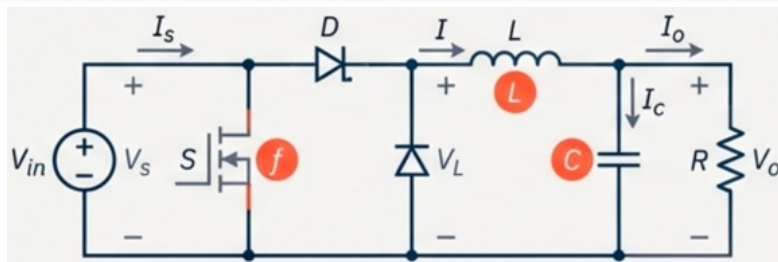
Generative Design: Deep Symbolic Optimization (DSO)



Token Sequence

Token Tree

Generated Circuit Schematic



Mechanism:

- Recurrent Neural Network (RNN) generates structure,
- Reinforcement learning (RL) assigns reward based on efficiency.

Component Sizing - Limits of Analytical Equations

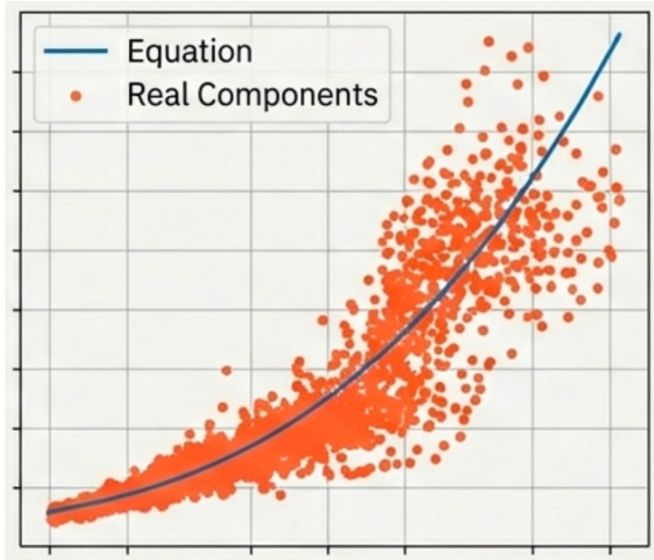
Physics Driven Formulas ← VERSUS → Data-Driven Reality

Standard FOM: $R_{\text{dson}} \times Q_g$

Assumption: Linear relationship

Unknown: Cost, pkg parasitics, non-linear voltage dependencies

STATISTICAL EXAMPLE



Machine Learned (ML)

FOM: Learned from thousands of parts

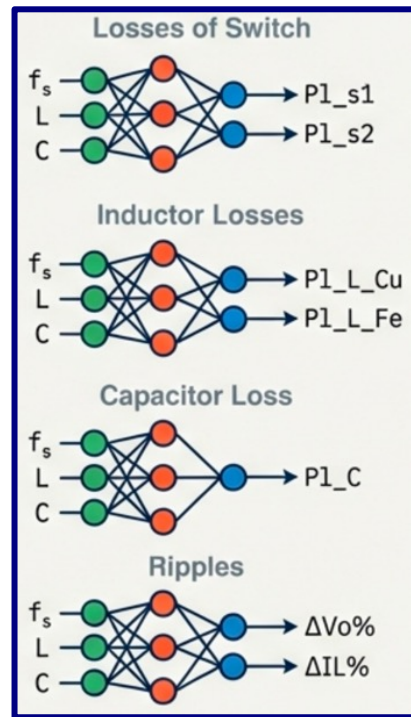
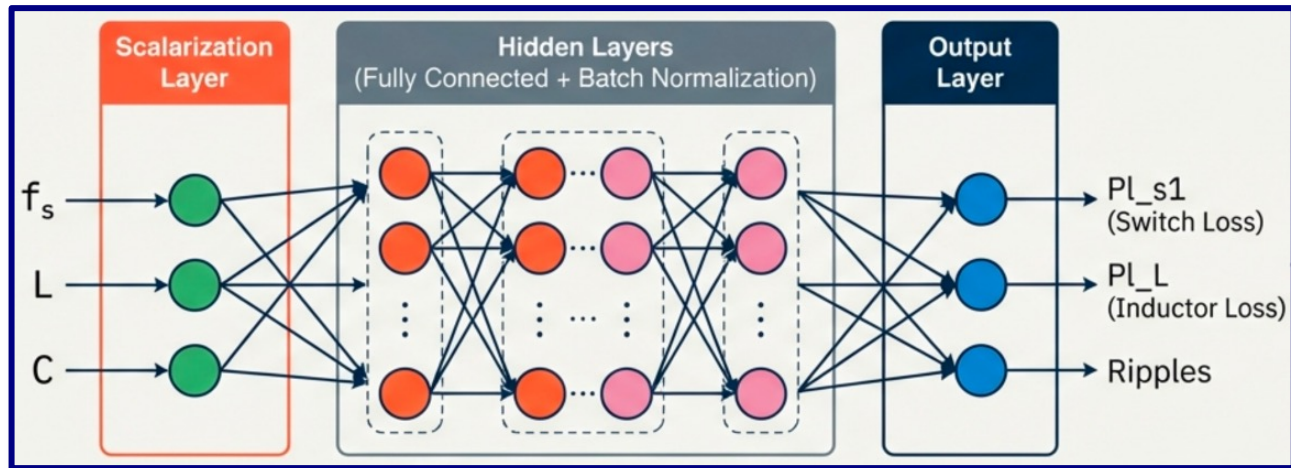
Reality: Captures non-linearities and commercial constraints

Example: SiC MOSFET R_{dson} at high voltage, high temperature is well predicted compared to SI MOSFETs that show inaccuracies

Verification – Surrogate Modeling (ANN)

*Bypassing the
Simulation Bottleneck*

Data-Driven Modeling



Completes the Integrated AI/ML Design Workflow

Current Challenges & Technical Hurdles

DATA Quality

Garbage In, Garbage Out. Models are limited by datasheet accuracy and missing mechanical data



Hallucination

LLMs can suggest physically plausible but incorrect circuits. Requires RAG grounding.



Physics-Informed AI

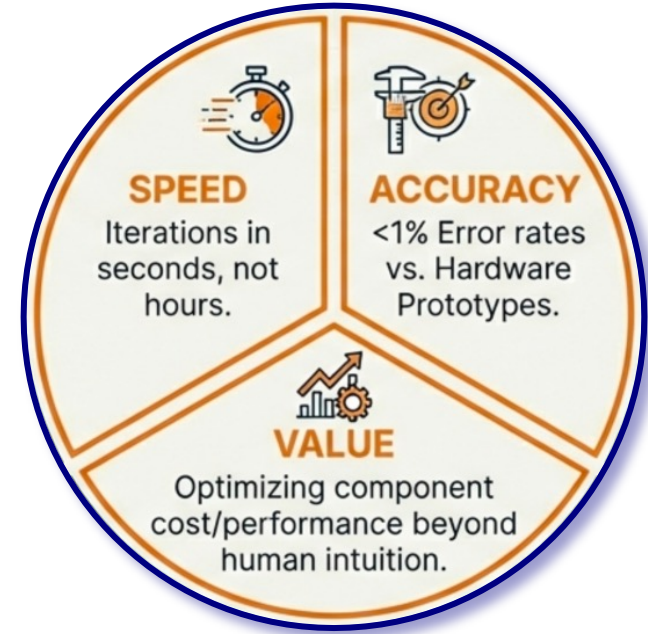
Need to integrate electrical and thermal laws into loss functions to prevent "impossible" circuit generation.

The Future Is In Agentic AI – Multiple Agents

Agentic AI refers to autonomous systems that act as agents to achieve specific goals with limited human supervision, moving beyond passive, prompt-based generative AI.

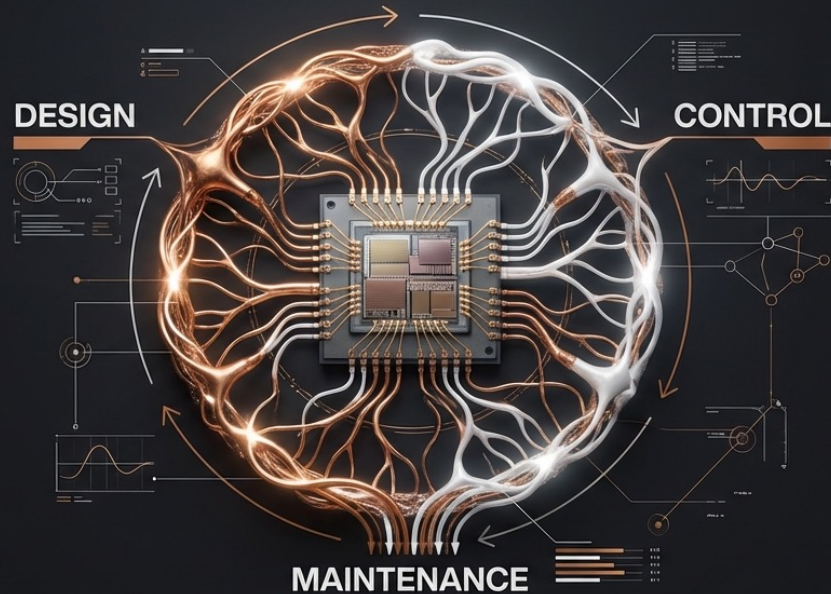
These systems use reasoning, planning, and tools to independently execute multi-step workflows, adapt to changing environments, and make context-aware decisions

AI replaces the calculation loop, freeing the engineer to focus on architecture and innovation.



QUESTIONS

The Era of the Cognitive Converter



Embrace the data-driven perspective to unlock efficiency, reliability, and speed.

APPENDICES

APPENDIX A: ANN Structure

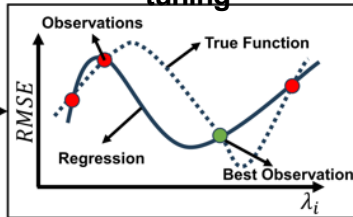
Hyperparameters (λ):

Number of hidden layers: [1,2,3]

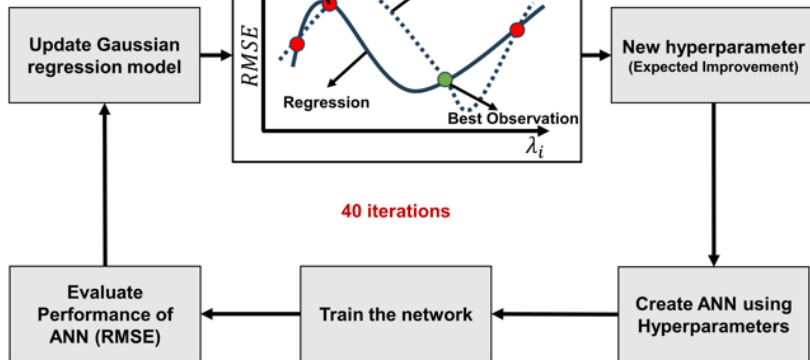
Number of nodes per layer: [1,...,300]

Activation functions: [Tanh, ReLU, Sigmoid]

Bayesian hyperparameter tuning

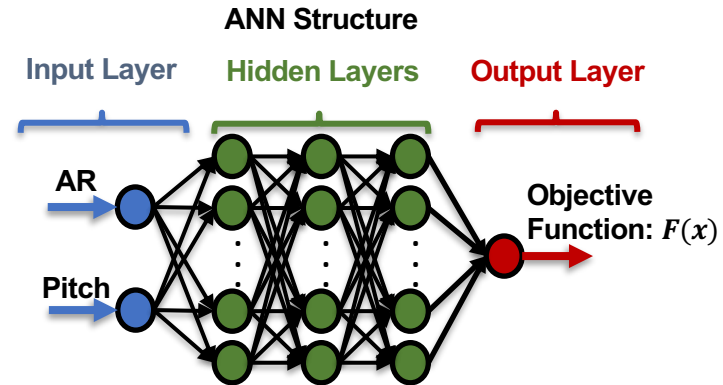


40 iterations

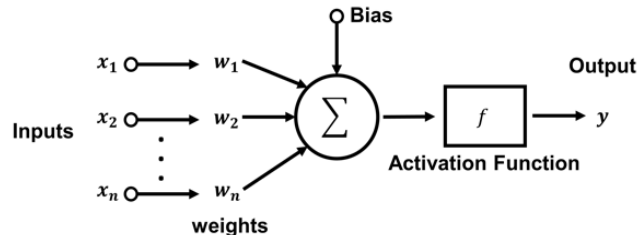


$[\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5]$

Number of nodes for layers 1-3
Number of Hidden layer
Activation Function



Activation functions in ANN



Sigmoid	$g(z) = \frac{1}{1 + e^{-z}}$
Tanh	$g(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$
ReLU	$g(z) = \max(0, z)$

APPENDIX B – Glossary

- Reinforcement learning (RL) is a machine learning technique where an AI "agent" learns to make decisions by interacting with an environment, using trial-and-error to achieve a goal
- A Recurrent Neural Network (RNN) is a type of deep learning model designed to process sequential data (text, audio, time-series) by using internal memory to retain information from previous inputs. Unlike traditional feedforward networks, RNNs have loops that feed output back into the network, enabling them to understand context and temporal dependencies.
- Design Space Optimization (DSO.ai™), a Synopsys technology using reinforcement learning to autonomously explore vast chip design possibilities, improving power, performance, and area (PPA) faster than manual methods.
- Retrieval-Augmented Generation (RAG) is a framework that improves Large Language Model (LLM) accuracy and relevance by fetching data from external, trusted sources—such as document databases, intranets, or web searches—before generating a response. It bridges the gap between static training data and real-time information, reducing hallucinations by grounding answers in retrieved context.
- A Large Language Model (LLM) is a type of AI designed to understand, process, and generate human-like text by analyzing massive datasets using deep learning, specifically transformer architectures. LLMs, such as GPT or BERT, learn to predict subsequent words in a sentence, allowing them to summarize, translate, and answer questions.

APPENDIX C – Suggested Reading

Using ChatGPT for the Knowledge and Design of Power Electronics Converters — Quanrui Liu, Xibo Yuan, Sergio Busquets-Monge (Universitat Politècnica de Catalunya, Barcelona, Spain), Jiang Feng, Yonglei Zhang, Kai Wang (China University of Mining and Technology, Xuzhou, China), *2024 IEEE 9th Southern Power Electronics Conference (SPEC)*, Brisbane, Australia, 2024, pp. 1-6, doi: 10.1109/SPEC62217.2024.10893269.

Machine Learning-based Component Figures of Merit and Models for DC-DC Converter Design — Skye Reese, Thomas Byrd (Lockheed Martin), John Haddon (Lockheed Martin), Dragan Maksimovic (University of Colorado Boulder), *2022 IEEE Design Methodologies Conference (DMC)*, Bath, United Kingdom, 2022, pp. 1-6, doi: 10.1109/DMC55175.2022.9906474

Deep Symbolic Optimization for Electric Component Sizing in Fixed Topology Power Converters — Ruben Glatt, Felipe Leno da Silva, Can Huang (Lawrence Livermore National Laboratory), Van-Hai Bui, Mengqi Wang, Fangyuan Chang, Yi Lu Murphey, Wencong Su (University of Michigan, Dearborn), Lingxiao Xue (Oak Ridge National Laboratory), *AAAI 2022 Workshop on AI for Design and Manufacturing (ADAM)*, Vancouver, Canada, Feb 28, 2022, (Published: 23 May 2023), LLNL-CONF-828961

An Approach to DC-DC Converter Optimization Using Machine Learning-based Component Models - Skye Reese, Dragan Maksimovic, *2022 IEEE 23rd Workshop on Control and Modeling for Power Electronics (COMPEL)*, Tel Aviv, Israel, 2022, pp. 1-8, doi: 10.1109/COMPEL53829.2022.9830017.

Artificial Neural Networks-Based Multi-Objective Design Methodology for Wide-Bandgap Power Electronics Converters - Rajesh Rajamony, Sheng Wang, Wenlong Ming, Gerardo Calderon-Lopez, Ingo Ludtke, *2022 IEEE 23rd Workshop on Control and Modeling for Power Electronics (COMPEL)*, Tel Aviv, Israel, 2022, pp. 1-8, doi: 10.1109/COMPEL53829.2022.9830017

Artificial-Intelligence-Based Design for Circuit Parameters of Power Converters - Xinze Li, Xin Zhang, Fanfan Lin, Frede Blaabjerg, *IEEE Transactions on Industrial Electronics*, vol. 69, no. 11, pp. 11144-11155, Nov. 2022, doi: 10.1109/TIE.2021.3088377

An Intelligent System for Automatic Selection of DC-DC Converter Topology with Optimal Design - Song Wang, Yi Lu Murphey, Wencong Su, Mengqi Wang, Van-Hai Bui, Fangyuan Chang, Can Huang, Lingxiao Xue, Ruben Glatt, *AAAI Workshop*, Denver, CO, March 1-4, 2022 LLNL-CONF-828953